CS24305 — THEORY (3.0 CR) | MODULE III NOTES | BIT MESRA | NEP SCHEME

Module 3: MAC Sublayer & LANs

Pure & Slotted ALOHA Proofs, CSMA/CD Minimum Frame Derivation, Exponential Backoff, Ethernet & Wi-Fi

Module III: Medium Access Control (MAC), Local Area Networks & Wireless LANs

1. Channel Allocation Problem: Static FDM/TDM vs. Dynamic Random Access
2. Pure ALOHA Protocol Mathematical Throughput Derivation ($S = Ge^{-2G} \implies 18.4\%$)
3. Slotted ALOHA Protocol Mathematical Throughput Derivation ($S = Ge^{-G} \implies 36.8\%$)
4. Carrier Sense Multiple Access (CSMA): 1-Persistent, Non-Persistent, p-Persistent
5. CSMA with Collision Detection (CSMA/CD — IEEE 802.3 Ethernet Standard)
6. Minimum Frame Size Derivation: $L_{\min} = 2 \times R \times \frac{d}{v}$ in Ethernet
7. Truncated Binary Exponential Backoff Algorithm ($0 \le k < 2^i$)
8. Standard Ethernet IEEE 802.3 Frame Structure & Physical MAC 48-Bit Addresses
9. Fast Ethernet (100 Mbps), Gigabit Ethernet (1 Gbps) & 10-Gigabit Standards
10. Wireless LANs (IEEE 802.11 Wi-Fi Architecture & CSMA/CA with RTS/CTS)
11. The Hidden Station & Exposed Station Problems with Virtual Carrier Sensing (NAV)
12. Comprehensive Solved BIT Mesra & GATE Exam Question Bank (8 Questions)

Topic 2 & 3: ALOHA Protocols & Poisson Throughput Proofs

1. Pure ALOHA Throughput Derivation: Let G be the total offered traffic load (frames per frame transmission time T). - Vulnerable Period in Pure ALOHA $= 2 \times T$ (any frame generated in $(t-T, t+T)$ collides). - Under Poisson distribution, probability of zero other frame arrivals is $P(0) = e^{-2G}$. - Throughput:

$$S = G \cdot e^{-2G}$$

- Maximum Throughput occurs at $\frac{dS}{dG} = e^{-2G}(1-2G) = 0 \implies G = 0.5$:

$$S_{\max} = 0.5 \cdot e^{-1} = \frac{1}{2e} \approx \mathbf{0.184} \quad (18.4\%)$$

2. Slotted ALOHA Throughput Derivation: - Frames can only be transmitted at discrete slot boundaries. - Vulnerable Period $= 1 \times T$. - Throughput:

$$S = G \cdot e^{-G}$$

- Maximum Throughput occurs at $G = 1.0$:

$$S_{\max} = 1.0 \cdot e^{-1} = \frac{1}{e} \approx \mathbf{0.368} \quad (36.8\%)$$

Topic 5 & 6: CSMA/CD & Minimum Frame Size Equation

In **CSMA/CD (Carrier Sense Multiple Access with Collision Detection)**, a transmitting station continuously listens to the physical cable while transmitting.

The Fundamental CSMA/CD Minimum Frame Size Rule

To guarantee that a station detects a collision before it finishes transmitting its frame, the transmission time T_t must be at least twice the maximum round-trip propagation delay $2 \times T_p$:

$$T_t \geq 2 \times T_p \implies \frac{L_{\min}}{B} \geq 2 \times \frac{d}{v} \implies L_{\min} = 2 \times B \times \frac{d}{v}$$

Where $L_{\min}$ is the minimum frame size in bits, B is the bandwidth in bps, d is the maximum network length in meters, and v is the signal propagation velocity in the cable ($\approx 2 \times 10^8 \text{ m/s}$).

🏛 **Step-by-Step Solved Problem: Classical 10 Mbps Ethernet 64-Byte Minimum Frame Size**

For standard 10 Mbps Ethernet with maximum collision domain distance 2500 meters and 4 repeaters:

$$2T_p \approx 51.2 \ \mu s$$

$$L_{\min} = 10 \times 10^6 \text{ bps} \times 51.2 \times 10^{-6} \text{ s} = 512 \text{ bits} = \mathbf{64} \text{ bytes}$$

Frames smaller than 64 bytes are padded with zeroes up to 64 bytes (Runt Frames < 64 bytes are discarded as collisions).

Topic 10 & 11: Wireless LANs (IEEE 802.11) & The Hidden Station Problem

Wireless Phenomenon	Geometric Problem Topology	CSMA/CA RTS/CTS Solution
1. Hidden Station Problem	Station A and Station C can both transmit to Base Station B, but A and C are out of range of each other. If both sense carrier simultaneously, they detect an idle channel and transmit, causing severe collisions at B.	**RTS/CTS Handshake:** A sends `RTS` (Request to Send). B broadcasts `CTS` (Clear to Send). C overhears B's `CTS` and sets its **Network Allocation Vector (NAV)** to defer transmission.
2. Exposed Station Problem	Station B is transmitting to A. Station C wants to transmit to D. C senses the medium, detects B's transmission, and wrongly defers, even though $C \to D$ would not interfere with $B \to A$.	Resolved by spatial channel reuse and directional antennas.

🏛 Top BIT Mesra Exam Questions & Answers (Module III)

Q1. Explain the Truncated Binary Exponential Backoff Algorithm used in Ethernet. (8 Marks)

After i collisions for a frame:
1. The station chooses a random integer k from the uniform range $0 \leq k < 2^c$, where $c = \min(i, 10)$.
2. The station waits $k \times \text{SlotTime}$ ($k \times 51.2 \ \mu s$ in 10 Mbps Ethernet) before attempting retransmission.
3. After 10 collisions, the backoff interval freezes at $2^{10} - 1 = 1023$ slots.
4. If the frame experiences 16 consecutive collisions ($i = 16$), the transmission is aborted and an error is reported to the upper layer.